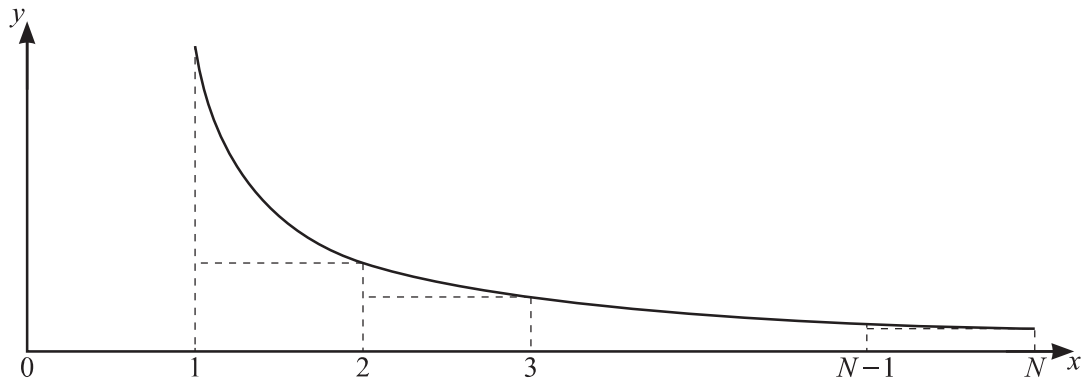




The diagram shows the curve $y = \frac{x}{2x^2 - 1}$ for $x \geq 1$, together with a set of $N - 1$ rectangles of unit width.

- (a) By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^N \frac{r}{2r^2-1} < \frac{1}{4} \ln(2N^2-1) + 1. \quad [7]$$

[illegible]

[illegible]